

A130 – The Fine-Structure Constant: α as Another Notation for ξ

Johann Pascher

July 2026

Contents

.1	The Thesis: In Natural Units $\alpha = 1$	1
.2	How $\alpha = 1$ Is Derived	1
.3	What Remains: ξ and One Scale	2
.4	Why the Measured Value Is Irrelevant	2
.5	Why the Circularity Is Not a Weakness	2
.6	The Scale E_0 as Entry Point	3
.7	The Two Paths Also Confirm K_{frak}	3
.8	A Coincidence That Must Be Resisted	4
.9	Verification Script	4
.10	Open Questions	4
.11	Supersedes	5
.12	Use of AI Tools	5

.1 The Thesis: In Natural Units $\alpha = 1$

[K] The strongest statement about the fine structure constant is the simplest: in natural units

$$\alpha = 1.$$

The famous value $1/137.036$ is *not* a mystery of nature, but the bookkeeping that arises when converting to SI units – just as $c = 3 \cdot 10^8$ is the bookkeeping of the metre-second choice.

This is not an invention of this theory. Setting α to one is a *known* unit convention: one normalises the charge so that $e^2 = 4\pi$ (rationalised Heaviside-Lorentz units with $\varepsilon_0 = \mu_0 = 1$). In these units $\alpha = 1$ by construction. The theory adopts this convention; it does not re-justify it.

But the reason is different from $c, \hbar, k_B$. For $c, \hbar, k_B$, the conversion factors become one because time-mass duality identifies their dimensions (A085). α is dimensionless; this justification does *not* apply here. The convention $\alpha = 1$ rests on the electromagnetic charge normalisation – a separate reason having nothing to do with time-mass duality and therefore handled here separately, not under the natural-units topic.

.2 How $\alpha = 1$ Is Derived

[B] The derivation runs through the charge normalisation and Maxwell duality, not through an axiom. In SI:

$$\alpha = \frac{e^2}{4\pi\varepsilon_0\hbar c}.$$

With $\hbar = c = 1$ and the natural choice $\epsilon_0 = 1$, the charge normalisation $e^2 = 4\pi$ follows from $\alpha = 1$. The Maxwell relation $c^2 = 1/(\epsilon_0\mu_0)$ with $c = 1$ and $\epsilon_0 = 1$ then gives $\mu_0 = 1$. Substituting confirms:

$$\alpha = \frac{e^2}{4\pi\epsilon_0\hbar c} = \frac{4\pi}{4\pi \cdot 1 \cdot 1 \cdot 1} = 1.$$

[K] The core is electromagnetic duality $\frac{1}{\epsilon_0 c} = \mu_0$, which is perfectly satisfied in natural units ($1 = 1$). $\alpha = 1$ is the expression of this balance between electric (ϵ_0) and magnetic (μ_0) coupling.

Honest assessment. The step itself is standard: the charge normalisation $e^2 = 4\pi$ is the rationalised convention in which $\alpha = 1$ comes out. What is non-standard is the *interpretation*: that the SI value $1/137$ is thereby not a fundamental mystery of the coupling but measures the historical charge unit. Not everyone shares this interpretation – the classical Duff position holds α to be absolute on account of its dimensionlessness. The theory's actual contribution lies not here, however, but in the next section: that the SI value not only measures charge units but, via $\alpha = \xi E_0^2$, carries the geometric parameter ξ .

This removes the mystification. The question "why exactly $1/137$?" is a question for the SI system, not for nature – of the same kind as "why exactly 299,792,458 metres per second?". The numerical value measures the historical charge unit, not a property of the coupling itself.

.3 What Remains: ξ and One Scale

[K] When all constants are one in natural units, the only genuine content remaining is the single dimensionless parameter ξ and one scale. The SI value of α carries precisely these two quantities:

$$\alpha_{\text{SI}} = \xi \cdot \left(\frac{E_0}{1 \text{ MeV}} \right)^2.$$

Here α is no longer the native one but its SI dress. In this dress one sees that α and ξ encode the same quantity, connected by the scale E_0 : ξ is the geometric reading (packing ratio, A020), α_{SI} the electromagnetic one. Two sides of one coin, one degree of freedom – just as mass, energy, temperature, and inverse time are one quantity in four dresses (A085).

.4 Why the Measured Value Is Irrelevant

[K] The theory works with ratios (A080). Ratios are unit-free and depend only on the geometry. The absolute value of $\alpha = 1/137.036$ – is an absolute value and therefore belongs to the bookkeeping layer, not the core.

[K] The *structural relations* at issue are already known and confirmed in SI units and in the Standard Model. The theory does not invent them, it orders them: $\alpha = \xi E_0^2$ connects known quantities. What it asserts is the *structure* of this connection – that α and ξ are the same degree of freedom – not a new numerical value for α .

[B] It follows how the numbers in the following sections are to be read: *not* as a test the theory passes or fails. They show the precision with which the bookkeeping reproduces the SI value – a statement about the bridge constant, not about the structure. As long as one works with ratios, they are irrelevant.

[STIPULATION] The test of the theory lies elsewhere. α is not one of the four prediction quantities of the series (A010); those are ratios and structural statements (m_{τ} , the ξ -order of the CHSH deviation, D_f , L_0). Whether α approaches $1/137$ to three or five digits does not test the theory – it tests the SI conversion.

.5 Why the Circularity Is Not a Weakness

[STIPULATION] As soon as one works in SI dress, the system is circular, and rightly so. Every system that fixes its own constants is circular – the only question is where one begins. (In natural units the question does not arise at all: there $\alpha = 1$ and nothing to determine.)

One can start from ξ (geometric perspective): then ξ is primary and $\alpha = \xi E_0^2$ derived. One can start from α (electromagnetic perspective): then α is primary and $\xi = \alpha / E_0^2$ derived. Both routes give the same values for all other constants. They are mathematically equivalent.

[B] The question “can α be derived from ξ without circularity?” is therefore wrongly posed. In a closed system there is no exterior from which one could derive. There is an entry point one declares, and the rest that follows from it. Classical physics conceals this by placing several constants side by side as “fundamental”; this theory makes explicit that only one entry point is needed.

A mistake would still be possible here, namely presenting the entry point as a derivation from nothing. Whoever substitutes $E_0 = \sqrt{\alpha/\xi}$ and then announces “exactly the measured value of α ” confuses reading consistency with proof. The E_0 determination avoids exactly this mistake, because it does *not* run via $\sqrt{\alpha/\xi}$: the exact E_0 is reached by Route 2 independently of α (previous section), and the agreement of the two α -free routes makes the α point over-determined rather than merely consistent. The circularity that always threatens in a closed system is therefore broken at this point by two separate approaches.

.6 The Scale E_0 as Entry Point

[K] What separates ξ from α is only the scale E_0 . For it there are *two mutually independent* routes, both not using α and confirming each other.

[K] Route 1 (empirical). The geometric mean of the two lightest lepton masses,

$$E_0 = \sqrt{m_e m_\mu} = 7.348 \text{ MeV},$$

gives when substituted $\alpha = \xi \cdot (7.348)^2 = 1/138.9$, thus -1.35% from the measured value.

[B] Route 2 (geometric). A construction from ξ and m_μ alone,

$$E_0^2 = 4\sqrt{2} \frac{m_\mu}{\xi p},$$

gives $E_0 = 7.398 \text{ MeV}$ – *without* using the geometric mean, *without* m_e , and *without* back-calculation from α . Substituted, this hits α to $5 \cdot 10^{-6}$.

[K] Decisive is that the two routes *meet* without knowing each other. Their ratio is

$$\frac{7.398}{7.348} = 1.00682 = K_{\text{frak}}^{-1/2} \quad \text{to } 9 \cdot 10^{-5},$$

with the same fractal factor $K_{\text{frak}} = 1 - 100\xi$ derived in A040 and acting in the corpus at an entirely different point. Since E_0 enters α quadratically, the $K_{\text{frak}}^{1/2}$ bridge at E_0 -level corresponds to a full K_{frak} correction at α -level. Two separate derivations thus meet to $8 \cdot 10^{-5}$, and the depth of the correction is not free but structurally fixed.

[K] The α point is thereby *not a calibration but over-determined*. The exact $E_0 = 7.398$ is not “formed backwards from α ” but reached forward by Route 2; and that precisely the derived K_{frak} closes the gap between the two routes is a cross-sector consistency, not a fit. Because Route 2 does not use m_e , the agreement with Route 1 is even a genuine statement about m_e : the remaining residual of $8 \cdot 10^{-5}$ is too large for mass measurement noise and is thus a test of the pole mass against the geometry.

.7 The Two Paths Also Confirm K_{frak}

[K] The over-determination says more than has been exploited so far. Neither Route 1 nor Route 2 uses K_{frak} anywhere: Route 1 is the geometric mean of two measured masses, Route 2 a construction from ξ and m_μ . Their ratio is therefore a *measurement* of K_{frak} that does not presuppose the factor:

$$K_{\text{frak}} = \left(\frac{E_0^{(2)}}{E_0^{(1)}} \right)^{-2} = \left(\frac{7.398}{7.348} \right)^{-2} = 0.9862.$$

[K] This value read from the two routes hits the factor independently derived in A040:

$$K_{\text{frak}} = 1 - 100\xi = \frac{74}{75} = 0.98667$$

to $9 \cdot 10^{-5}$. This is remarkable because the two origins have nothing to do with each other: A040 obtains K_{frak} from the cumulative RG run and the consistency condition on the effective fractal dimension; the α sector obtains *the same* value from the difference between two energy-scale routes. A factor originating from an RG argument that simultaneously closes the gap of two independent E_0 determinations is not freely chosen; it is fixed from two sides.

[B] K_{frak} thus has a *second witness* here. The α sector confirms not only the value of E_0 and α , but via the route difference also that $K_{\text{frak}} = 74/75$ has the correct value – without inserting this factor anywhere. What in A040 is a derived constant is here additionally *measured*.

.8 A Coincidence That Must Be Resisted

[K] It is tempting to search for a *more elegant* expression for E_0 than the mass route, and there is a striking one: Euler's number squared,

$$e^2 = 7.389 \text{ MeV } (?), \quad \alpha = \xi \cdot (7.389)^2 = \frac{1}{137.37} \quad (-0.24 \%),$$

which hits the SI value even better than the mass route. This expression should be *resisted*, for a principled reason: it equates a pure number (Euler's number) with an energy in the arbitrary unit MeV. One MeV is a human convention; no pure number can fix a physical scale *in* MeV without the agreement depending on the unit choice. In a different energy unit a different number would appear, and the coincidence would vanish.

[K] This is exactly the pattern the corpus warns against elsewhere (A105): a number fitted backwards to an elegant expression without forward content. That e^2 hits closer than $\sqrt{m_e m_\mu}$ does *not* make it the better candidate – the mass route carries a physical reason (the lepton masses), the Euler coincidence carries none. It is recorded here only to explicitly *not* book it as an anchor. The elegant expression is not needed anyway: the exact E_0 is already hit by Route 2 with forward content, while e^2 merely holds a number to a unit.

.9 Verification Script

- The equivalence of both perspectives ($\alpha = \xi E_0^2$ and $\xi = \alpha/E_0^2$ deliver identical values); the α -free harmonic route with $\kappa = 4$; the two numerical routes; and the explicit check that substituting $\sqrt{\alpha/\xi}$ is a consistency and not an independent proof: `python/A_Serie_Skripte/a130_alpha_kette.py`

.10 Open Questions

First, the exact E_0 is α -free and over-determined – not open. The exact $E_0 = 7.398$ is reached by Route 2 from ξ and m_{μ} without α and without m_{e_i} ; it agrees with the mass route (Route 1) up to the derived factor $K_{\text{frak}}^{-1/2}$. The earlier classification that the exact value was formed only backwards from α is thereby superseded. The Euler coincidence $E_0 = e^2$ remains explicitly *not* a derivation and is not needed.

Second, the absolute anchor remains tied to a measured mass. Route 2 fixes E_0 from ξ and m_{μ} ; the absolute scale therefore enters via *one* measured mass anchor. A determination of the exact E_0 from purely dimensionless geometry without any mass anchor does not exist – but this is the general structure of the SI bridge (A080), not a deficiency specific to the α sector. Since the theory tests via ratios, the remaining residual carries no weight for its evaluation.

Third, a lower bound applies. The relative description $\alpha \leftrightarrow \xi$ is secured down to the sub-Planck bound $L_0 = \xi \ell_P$ (A210); below this the theory makes no statement.

.11 Supersedes

This document subsumes: Doc. 011 (fine structure constant), Doc. 043 (resolution of constants), Doc. 044, Doc. 101 (circularity of constants, core thesis: one degree of freedom, ξ and α equivalent). Incorporated: P40.

.12 Use of AI Tools

This document was created with the assistance of AI tools. The questions, stipulations, and all substantive decisions come from the author; the tools formulate, compute, and create verification scripts. Responsibility for content and errors rests entirely with the author. Full wording of the rules in A010.